CSS Measurement Units

Maheesha Fernando
Bsc (Hons) in Computer Science

MEASUREMENT UNITS

In CSS, measurement units are used to control how big or small elements appear on a web page. These units decide spacing, text size, and layout behavior.

CSS units define size, spacing, and layout
Used in:

- width, height
- margin, padding
- font-size
- positioning

TYPES OF CSS UNITS

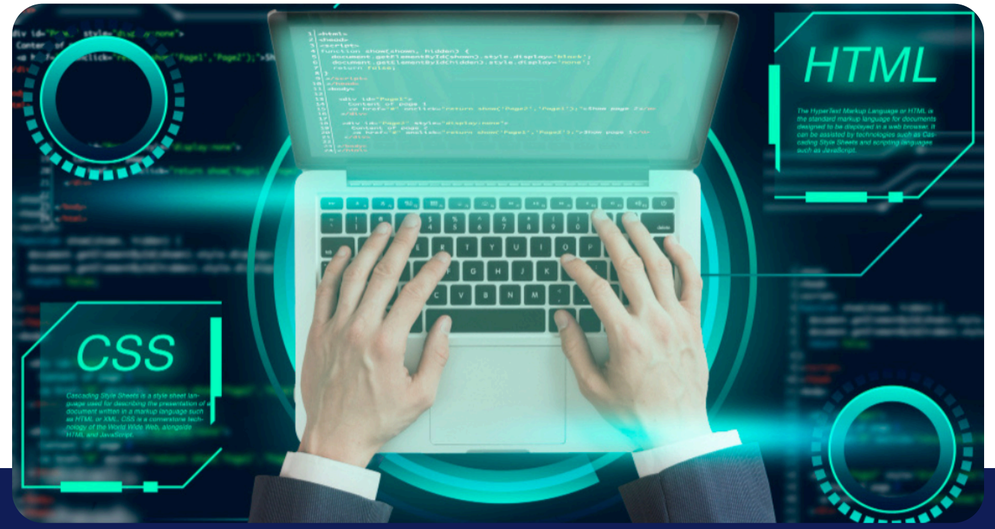
- Absolute Units
- Relative Units

ABSOLUTE UNITS

Absolute units have a fixed size. No matter the screen or device, their value stays the same. That’s why they are not very responsive.

Absolute Units

- px – pixels
- cm – centimeters
- mm – millimeters
- in – inches



Absolute Units – Example Code

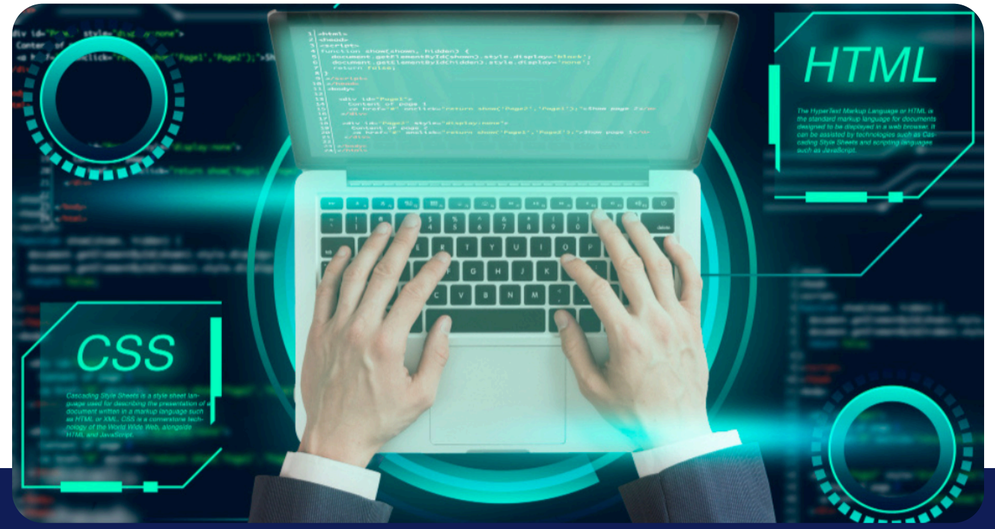
```html
<head>
  <style>
    .box {
      width: 300px;
      height: 150px;
      background-color: lightblue;
      margin: 20px;
    }

    .text {
      font-size: 16px;
    }
  </style>
</head>
<body>

  <div class="box"></div>
  <p class="text">This text uses px unit</p>

</body>
```

RELATIVE UNITS

Relative units change based on screen size or parent element size. This makes layouts flexible and responsive.

Relative Units

- % – relative to parent element
- em – relative to parent font-size
- rem – relative to root (html) font-size
- vw – viewport width
- vh – viewport height



Absolute Units – Percentage (%) Example

```
.container {
  width: 100%;
}

.child {
  width: 50%;
  background-color: lightcoral;
}
```

Realtive Units – Percentage (%) Example

```
.container {
  width: 100%;
}

.child {
  width: 50%;
  background-color: lightcoral;
}
```

Here, the child div takes 50% of its parent’s width. If the parent resizes, the child resizes automatically.

Relative Units – em vs rem Example

```
html {
  font-size: 16px;
}

.parent {
  font-size: 20px;
}

.child-em {
  font-size: 1.5em; /* 30px */
}

.child-rem {
  font-size: 1.5rem; /* 24px */
}
```

em depends on the parent's font size, but **rem** depends on the root element. That's why rem is more predictable.

Relative Units – vw and vh Example

```
.hero {
  width: 100vw;
  height: 100vh;
  background-color: lightgreen;
}
```

- **vw** and **vh** depend on the browser window size.
- **100vh** means full screen height.

When to Use Absolute Units

- ✓ Borders
- ✓ Icons
- ✓ Small fixed UI elements

When to Use Relative Units

- ✓ Font sizes
- ✓ Layout widths
- ✓ Responsive designs